

Gabrielle Pang

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EDUCATION

NANYANG POLYTECHNIC	2023 – 2026
<i>Diploma in Robotics and Mechatronics, School of Engineering</i>	<i>Singapore</i>
HOLY INNOCENTS' HIGH SCHOOL	2018 – 2022
<i>GCE O-Level</i>	<i>Singapore</i>

EXPERIENCE

ROBOT BARISTA ARM AUTOMATION – Overseas Industrial Training Programme	2025
<i>Team Project Robotic Systems & Automation</i>	<i>Korea</i>
- Applied GPIO-triggered control logic between Raspberry Pi and a robotic arm to execute precise and repeatable automated drink-making sequences. - Built and reviewed RoboDK simulations to validate motion paths, positioning and sequence flow before physical testing. - Programmed, tested and refined predefined robotic movements through repeated runs, visual inspection and technical troubleshooting.	

AI & TECHNICAL PROJECTS

CV AGENT – Telegram AI Productivity & Visual Reasoning Assistant	2026
<i>Personal AI Agent Project</i>	<i>Singapore</i>
- Engineered a multimodal AI agent integrating Qwen2.5-VL visual reasoning, Groq-powered language models, Gmail IMAP, SQLite and a Flask dashboard. - Implemented natural-language workflows for visual note extraction, email triage, reminders, application tracking, scheduled briefings and AI-generated summaries.	

AI SUPPORT TICKET TRIAGE AGENT – NLP Classification & Workflow Automation	2026
<i>Personal Machine Learning & Automation Project</i>	<i>Singapore</i>
- Built an end-to-end NLP triage agent using Python, TF-IDF and scikit-learn to classify ticket categories and priorities, achieving 95.9% and 90.0% accuracy on operational labels. - Automated escalation, human-review and lower-tier resolution workflows with PII redaction, saved Joblib models, email processing and a Streamlit monitoring dashboard.	

SMART IOT PEOPLE COUNTER – Smart IR Detection System	2025
<i>Final Year Project Developed for use in NYP Library</i>	<i>Singapore</i>
- Developed an edge-to-cloud people-counting system using ESP8266 microcontrollers and IR sensors to classify entry and exit movement while reducing false counts. - Built live occupancy monitoring through Flutter and web dashboards, connecting sensor nodes through a central WiFi communication hub. - Integrated AWS IoT Core, Lambda and DynamoDB for cloud ingestion, processing and storage; performed hardware testing, debugging and validation.	

LEADERSHIP & ACTIVITIES

GEO COUNCIL – NANYANG POLYTECHNIC	2025
<i>Member</i>	<i>Singapore</i>
Designed visual content for student events and incorporated feedback to maintain quality and consistency.	
FLOORBALL TEAM – HOLY INNOCENTS' HIGH SCHOOL	2018 – 2022
<i>CCA Team Member</i>	<i>Singapore</i>
Represented the school in B and C Division competitions; qualified for B Division Round 2 in 2022, the first time in school history.	

CERTIFICATIONS & ACHIEVEMENTS

- JPMorgan Quantitative Research Job Simulation – Forage (Jul 2026) • AI for MSME Advancement Training Programme – ASEAN Foundation (Apr 2026)
- KOPO Overseas Internship Programme – Korea Polytechnics, Changwon (Apr 2025; no expiry) • Smart Parking System selected for the NYP Open House showcase (2024)
- Floorball B Division – qualified for Round 2, first in school history (2022) • ActiveSG SFA C Division Plate & Bowl – 3rd Place (2018)

SKILLS & INTERESTS

AI & Machine Learning: Python, AI Agents, NLP, TF-IDF, scikit-learn, LLM/VLM Integration, Qwen2.5-VL, Groq API, OpenCV, Image Processing
Data, Cloud & Backend: Pandas, NumPy, Streamlit, Joblib, AWS IoT Core, Lambda, DynamoDB, Flask, SQLite, Firebase, API Integration
IoT & Robotics: ESP8266, Raspberry Pi, Arduino, GPIO, IR Sensors, WiFi Communication, RoboDK, Robotic Arms, Soldering, CAD Modelling
Professional: Testing, Debugging, Technical Documentation, Analytical Thinking, Team Collaboration, Adaptability | **Languages:** English (Fluent), Mandarin